HW 2

Wednesday, 15 February 2023 3:16 PM

Question 1

As shown in class, for a periodic dirac-comb w/ periodicity of $n = 0, \pm 1, \pm 2 \ldots$

$0 < x < a$, $V(x) = 0$: TISE : $\frac{-\hbar^2}{2m}\frac{d^2\psi}{dx^2} = E\psi$

$$\psi(x) = A\sin(kx) + B\cos(kx) \quad ; \quad k = \sqrt{\frac{2mE}{\hbar^2}}$$

Applying Bloch's Theorem : $\psi(x+a) = e^{ika}\left[A\sin(k(x+a)) + B\cos(k(x+a))\right]$; $K = \frac{2\pi n}{P \cdot a}$

for $-a < x < 0$

Plugging in initial conditions of : @ $x = 0$, $\psi_{0^-} = \psi_{0^+}$ (continuity)

& $\left.\frac{d\psi}{dx}\right|_{0^+} - \left.\frac{d\psi}{dx}\right|_{0^-} = \frac{2m\alpha}{\hbar^2}\psi(0)$ ["cusp"]

and eliminating for A and B:

$$\cos(Ka) = \cos(ka) + \frac{m\alpha a}{\hbar^2}\frac{\sin(ka)}{ka} \quad \ldots \text{ Eq 1}$$

Equation 1 can be solved numerically : $m = \alpha = a = \hbar = 1$

$$\therefore \cos(K) = \cos(k) + \frac{\sin(k)}{k} \quad ; \quad K = \frac{2\pi n}{P} \quad ; \quad k = \sqrt{2E}$$

(a) → Solving for the specific k values given K values for the first band ($n = 0, \pm 1, \pm 2, \pm 3, \pm 4, \pm 5$) P+1 non-interacting fermions we obtain the following table:

For $P = 10$; $n = P + 1 = 11$ particles

$P = 20$; $n = P + 1 = 21$ particles

; A band is defined when we fill all discrete energy levels between $1 < \cos(K) < -1$

	P =	10			
n	K	cos(K)	k	Energy	Discrete Band Energy
0	0.000	1.000	1.307	0.854	
1	0.628	0.809	1.447	1.047	
-1	-0.628	0.809	1.447	1.047	
2	1.257	0.309	1.803	2.581	
-2	-1.257	0.309	1.803	2.581	
3	1.885	-0.309	2.272	3.867	**33.4**
-3	-1.885	-0.309	2.272	3.867	
4	2.513	-0.809	2.781	3.867	
-4	-2.513	-0.809	2.781	3.867	
5	3.142	-1.000	3.142	4.935	
-5	-3.142	-1.000	3.142	4.935	

	P =	20			
n	K	cos(K)	k	Energy	Discrete Band Energy
0	0.000	1.000	1.307	**0.854**	
1	0.314	0.951	1.343	**0.902**	
-1	-0.314	0.951	1.343	**0.902**	
2	0.628	0.809	1.447	**1.047**	
-2	-0.628	0.809	1.447	**1.047**	
3	0.942	0.588	1.606	**1.290**	
-3	-0.942	0.588	1.606	**1.290**	
4	1.257	0.309	1.803	**1.625**	
-4	-1.257	0.309	1.803	**1.625**	
5	1.571	0.000	2.092	**2.188**	
-5	-1.571	0.000	2.092	**2.188**	**53.2**
6	1.885	-0.309	2.272	**2.581**	
-6	-1.885	-0.309	2.272	**2.581**	
7	2.199	-0.588	2.526	**3.190**	
-7	-2.199	-0.588	2.526	**3.190**	
8	2.513	-0.809	2.781	**3.867**	
-8	-2.513	-0.809	2.781	**3.867**	
9	2.827	-0.951	3.017	**4.551**	
-9	-2.827	-0.951	3.017	**4.551**	
10	3.142	-1.000	3.142	**4.935**	
-10	-3.142	-1.000	3.142	**4.935**	

(b) P=10, but first two bands are occupied

n	K	cos(K)	k	Energy	Discrete Band Energy
0	0.000	1.000	1.307	0.854	
1	0.628	0.809	1.447	1.047	
-1	-0.628	0.809	1.447	1.047	
2	1.257	0.309	1.803	2.581	
-2	-1.257	0.309	1.803	2.581	
3	1.885	-0.309	2.272	3.867	33.4
-3	-1.885	-0.309	2.272	3.867	
4	2.513	-0.809	2.781	3.867	
-4	-2.513	-0.809	2.781	3.867	
5	3.142	-1.000	3.142	4.935	
-5	-3.142	-1.000	3.142	4.935	
6	3.770	-0.809	4.051	8.205	
-6	-3.770	-0.809	4.051	8.205	
7	4.398	-0.309	4.619	10.668	
-7	-4.398	-0.309	4.619	10.668	
8	5.027	0.309	5.211	13.577	171.5
-8	-5.027	0.309	5.211	13.577	
9	5.655	0.809	5.806	16.855	
-9	-5.655	0.809	5.806	16.855	
10	6.2832	1.000	6.283	19.739	
-10	-6.283	1.000	6.283	19.739	

Total N = 21

(c) P=10, but first three bands are occupied.

	P =	10			
n	K	cos(K)	k	Energy	Discrete Band Energy
0	0.000	1.000	1.307	0.854	
1	0.628	0.809	1.447	1.047	
-1	-0.628	0.809	1.447	1.047	
2	1.257	0.309	1.803	2.581	
-2	-1.257	0.309	1.803	2.581	
3	1.885	-0.309	2.272	3.867	33.4
-3	-1.885	-0.309	2.272	3.867	
4	2.513	-0.809	2.781	3.867	
-4	-2.513	-0.809	2.781	3.867	
5	3.142	-1.000	3.142	4.935	
-5	-3.142	-1.000	3.142	4.935	
6	3.770	-0.809	4.051	8.205	
-6	-3.770	-0.809	4.051	8.205	
7	4.398	-0.309	4.619	10.668	
-7	-4.398	-0.309	4.619	10.668	
8	5.027	0.309	5.211	13.577	171.5
-8	-5.027	0.309	5.211	13.577	
9	5.655	0.809	5.806	16.855	
-9	-5.655	0.809	5.806	16.855	
10	6.2832	1.000	6.283	19.739	
-10	-6.283	1.000	6.283	19.739	
11	6.912	0.809	7.066	24.964	
-11	-6.912	0.809	7.066	24.964	
12	7.540	0.309	7.672	29.430	
-12	-7.540	0.309	7.672	29.430	
13	8.168	-0.309	8.286	34.329	
-13	-8.168	-0.309	8.286	34.329	517.0
14	8.796	-0.809	8.900	39.605	
-14	-8.796	-0.809	8.900	39.605	
15	9.425	-1.000	9.425	44.413	
-15	-9.425	-1.000	9.425	44.413	

Total N = 31

(d) As seen from the table below, the integral method (which was a trapezoid numerical integration)

(d) As seen from the table below, the integral method (which was a trapezoid numerical integration) are larger than energies calculated w/ discrete energy summation

P = 10

n	K	cos(K)	k	Energy	Discrete Band Energy	Integral Band Energy
0	0.000	1.000	1.307	0.854		
1	0.628	0.809	1.447	1.047		
-1	-0.628	0.809	1.447	1.047		
2	1.257	0.309	1.803	2.581		
-2	-1.257	0.309	1.803	2.581		
3	1.885	-0.309	2.272	3.867	33.4	36.8
-3	-1.885	-0.309	2.272	3.867		
4	2.513	-0.809	2.781	3.867		
-4	-2.513	-0.809	2.781	3.867		
5	3.142	-1.000	3.142	4.935		
-5	-3.142	-1.000	3.142	4.935		
6	3.770	-0.809	4.051	8.205		
-6	-3.770	-0.809	4.051	8.205		
7	4.398	-0.309	4.619	10.668		
-7	-4.398	-0.309	4.619	10.668		
8	5.027	0.309	5.211	13.577	171.5	174.9
-8	-5.027	0.309	5.211	13.577		
9	5.655	0.809	5.806	16.855		
-9	-5.655	0.809	5.806	16.855		
10	6.2832	1.000	6.283	19.739		
-10	-6.283	1.000	6.283	19.739		
11	6.912	0.809	7.066	24.964		
-11	-6.912	0.809	7.066	24.964		
12	7.540	0.309	7.672	29.430		
-12	-7.540	0.309	7.672	29.430		
13	8.168	-0.309	8.286	34.329	517.0	520.3
-13	-8.168	-0.309	8.286	34.329		
14	8.796	-0.809	8.900	39.605		
-14	-8.796	-0.809	8.900	39.605		
15	9.425	-1.000	9.425	44.413		
-15	-9.425	-1.000	9.425	44.413		

P = 20

n	K	cos(K)	k	Energy	Discrete Band Energy	Integral Band Energy	DFT Ground State Energy
0	0.000	1.000	1.307	**0.854**			
1	0.314	0.951	1.343	**0.902**			
-1	-0.314	0.951	1.343	**0.902**			
2	0.628	0.809	1.447	**1.047**			
-2	-0.628	0.809	1.447	**1.047**			
3	0.942	0.588	1.606	**1.290**			
-3	-0.942	0.588	1.606	**1.290**			
4	1.257	0.309	1.803	**1.625**			
-4	-1.257	0.309	1.803	**1.625**			
5	1.571	0.000	2.092	**2.188**			
-5	-1.571	0.000	2.092	**2.188**	53.2	55.7	59.1
6	1.885	-0.309	2.272	**2.581**			
-6	-1.885	-0.309	2.272	**2.581**			
7	2.199	-0.588	2.526	**3.190**			
-7	-2.199	-0.588	2.526	**3.190**			
8	2.513	-0.809	2.781	**3.867**			
-8	-2.513	-0.809	2.781	**3.867**			
9	2.827	-0.951	3.017	**4.551**			
-9	-2.827	-0.951	3.017	**4.551**			
10	3.142	-1.000	3.142	**4.935**			
-10	-3.142	-1.000	3.142	**4.935**			
11	3.456	-0.951	3.802	**7.228**			
-11	-3.456	-0.951	3.802	**7.228**			
12	3.770	-0.809	4.051	**8.205**			
-12	-3.770	-0.809	4.051	**8.205**			
13	4.084	-0.588	4.330	**9.374**			
-13	-4.084	-0.588	4.330	**9.374**			
14	4.398	-0.309	4.619	**10.668**			
-14	-4.398	-0.309	4.619	**10.668**			
15	4.712	0.000	4.913	**12.069**			
-15	-4.712	0.000	4.913	**12.069**	316.1	318.5	153.5
16	5.027	0.309	5.210	**13.572**			
-16	-5.027	0.309	5.210	**13.572**			
17	5.341	0.588	5.509	**15.175**			
-17	-5.341	0.588	5.509	**15.175**			
18	5.655	0.809	5.806	**16.855**			
-18	-5.655	0.809	5.806	**16.855**			
19	5.969	0.951	6.093	**18.562**			
-19	-5.969	0.951	6.093	**18.562**			
20	6.283	1.000	6.283185	**19.739**			
-20	-6.283	1.000	6.283185	**19.739**			

(e) DFT energy per unit cell for all cases:

For dirac comb, the DFT energy functional is given by: $E[\rho] = \frac{\pi^2\hbar^2}{6m}\int_{na}^{na} \rho^3\,dx + \int_{-na}^{na} V_{ext}(x)\rho\,dx$; n = some integer

$$\frac{\delta F}{\delta \rho} = 0 \Rightarrow \rho_{min}(x) = \left(\frac{2m}{\pi^2\hbar^2}(\mu - V_{ext}(x))\right)^{1/2}$$

∴ Solving for ground state energy: $\frac{E_{GS}}{P} = \frac{\pi^2\hbar^2}{6ma^2} N_{cell}^3 + \frac{\alpha}{a} N_{cell}$; $m = a = \hbar = \alpha = 1$; $N_{cell} = \frac{N}{P}$

$$\frac{E_{GS}}{P} = \frac{\pi^2}{6} N_{cell}^3 + N_{cell}$$

The table below shows the calculated values for DFT compared to those calculated via the integral method

P = 10

n	K	cos(K)	k	Energy	Discrete Band Energy	Integral Band Energy	DFT Ground State Energy	% Difference
0	0.000	1.000	1.307	0.854				
1	0.628	0.809	1.447	1.047				
-1	-0.628	0.809	1.447	1.047				
2	1.257	0.309	1.803	2.581				
-2	-1.257	0.309	1.803	2.581				
3	1.885	-0.309	2.272	3.867	33.4	36.8	32.9	9.0%
-3	-1.885	-0.309	2.272	3.867				
4	2.513	-0.809	2.781	3.867				
-4	-2.513	-0.809	2.781	3.867				
5	3.142	-1.000	3.142	4.935				
-5	-3.142	-1.000	3.142	4.935				
6	3.770	-0.809	4.051	8.205				
-6	-3.770	-0.809	4.051	8.205				
7	4.398	-0.309	4.619	10.668				
-7	-4.398	-0.309	4.619	10.668				
8	5.027	0.309	5.211	13.577	171.5	174.9	173.3	1.0%
-8	-5.027	0.309	5.211	13.577				
9	5.655	0.809	5.806	16.855				
-9	-5.655	0.809	5.806	16.855				
10	6.2832	1.000	6.283	19.739				
-10	-6.283	1.000	6.283	19.739				
11	6.912	0.809	7.066	24.964				
-11	-6.912	0.809	7.066	24.964				
12	7.540	0.309	7.672	29.430				
-12	-7.540	0.309	7.672	29.430				
13	8.168	-0.309	8.286	34.329	517.0	520.3	521.0	0.1%
-13	-8.168	-0.309	8.286	34.329				
14	8.796	-0.809	8.900	39.605				
-14	-8.796	-0.809	8.900	39.605				
15	9.425	-1.000	9.425	44.413				
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4	1.257	0.309	1.803	**1.625**				
-4	-1.257	0.309	1.803	**1.625**				
5	1.571	0.000	2.092	**2.188**				
-5	-1.571	0.000	2.092	**2.188**	53.2	55.7	59.1	4.5%
6	1.885	-0.309	2.272	**2.581**				
-6	-1.885	-0.309	2.272	**2.581**				
7	2.199	-0.588	2.526	**3.190**				
-7	-2.199	-0.588	2.526	**3.190**				
8	2.513	-0.809	2.781	**3.867**				
-8	-2.513	-0.809	2.781	**3.867**				
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-9	-2.827	-0.951	3.017	**4.551**				
10	3.142	-1.000	3.142	**4.935**				
-10	-3.142	-1.000	3.142	**4.935**				
11	3.456	-0.951	3.802	**7.228**				
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13	4.084	-0.588	4.330	**9.374**				
-13	-4.084	-0.588	4.330	**9.374**				
14	4.398	-0.309	4.619	**10.668**				
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15	4.712	0.000	4.913	**12.069**				
-15	-4.712	0.000	4.913	**12.069**	316.1	318.5	153.5	0.7%
16	5.027	0.309	5.210	**13.572**				
-16	-5.027	0.309	5.210	**13.572**				
17	5.341	0.588	5.509	**15.175**				
-17	-5.341	0.588	5.509	**15.175**				
18	5.655	0.809	5.806	**16.855**				
-18	-5.655	0.809	5.806	**16.855**				
19	5.969	0.951	6.093	**18.562**				
-19	-5.969	0.951	6.093	**18.562**				
20	6.283	1.000	6.283185	**19.739**				
-20	-6.283	1.000	6.283185	**19.739**				

The DFT values fall in between the discrete and and integral calculations and the difference b/w the integral band values decrease from P=10 to P=20. (i.e. DFT becomes more accurate)

Question 2

$V(x) = \alpha \sin\left(\frac{2\pi x}{a}\right)$ ← sin-wave potential

We know, $E[\rho] = \frac{\pi^2\hbar^2}{6m}\int_{-na}^{na} \rho^3\,dx + \int_{-na}^{na} V(x)\,\rho\,dx$;

Lagrangian, $F[\rho] = E[\rho] - \mu\left(\int_{-na}^{na} \rho\,dx - N\right)$

For $\alpha = 5$,

$V(x) = 5\sin\left(\frac{2\pi x}{L}\right)$ → period of 1

$$F[\rho] = \frac{\pi^2}{6}\int_0^1 \rho^3\,dx + \int_0^1 5\sin(2\pi x)\,\rho\,dx - \mu\left(\int_0^1 \rho\,dx - N\right)$$

Solving this,

$$\rho_{min}(x) = \left(\frac{2m}{\pi^2\hbar^2}\left(\mu - 5\sin\frac{2\pi x}{1}\right)\right)^{1/2}$$

$$\int_0^1 \rho_{min}(x)\,dx = N \quad \rightarrow \quad \text{Constraint}$$

$$\therefore \int_0^1 \frac{2}{\pi^2}\left(\mu - 5\sin(2\pi x)\right) = N$$

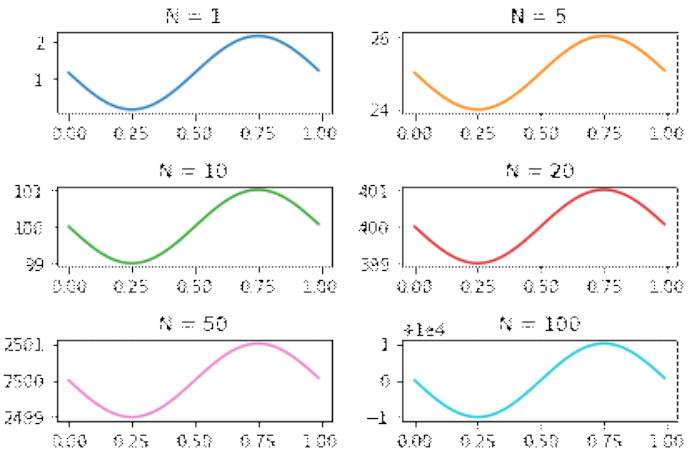
← Solving this numerically for μ for each N value

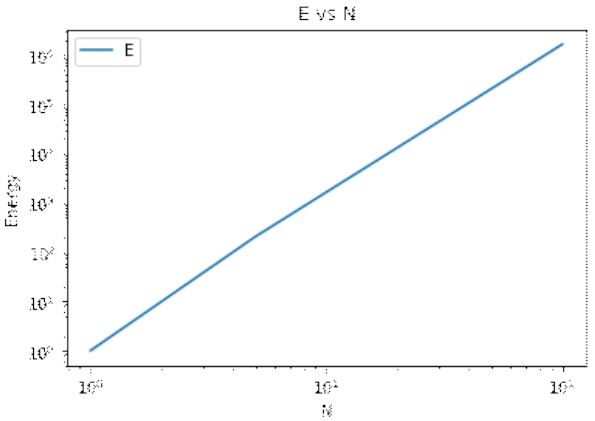
→ These μ values are then used to find ρ_{min}

↓

That is used to find $E[\rho_{min}]$ by evaluating the integral numerically

Thus, we can calculate ρ_{min} & then $E[\rho_{min}]$ ← These values are shown below.





↑ Graph above shows the periodic nature of the densities (plotted for $0 < x < 1$ to showcase the sin curve behaviour)

↑ Log-Log plot of E vs. N. This shows a linear relationship → ie. power law behaviour b/w E and N

